

Polynomial Time

Abstract Problems, Encodings, and the Class P

1 Polynomial Time

Central Idea

The study of NP-completeness begins by formalizing which problems are considered efficiently solvable. The standard mathematical notion of efficiency is **polynomial time**.

$$\text{running time} = \mathcal{O}(n^k)$$

for some constant k , where n is the input length.

Polynomial-time algorithms are generally regarded as tractable. This is partly a philosophical choice, not a purely mathematical fact, because an algorithm running in $\mathcal{O}(n^{100})$ time is technically polynomial but not practically useful.

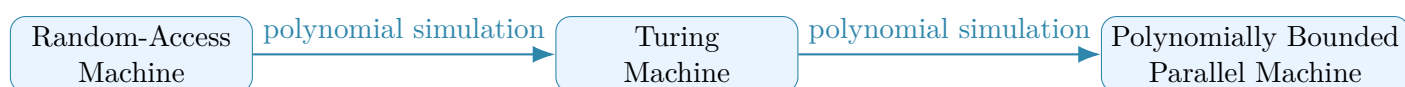
Why Polynomial Time Is Useful

Polynomial time is still the right boundary for complexity theory because:

1. Practical polynomial-time algorithms usually have small exponents.
2. Polynomial-time solvability is stable across reasonable models of computation.
3. Polynomial functions have strong closure properties under composition, addition, and multiplication.

1.1 Stability Across Models

For many reasonable models of computation, polynomial time does not depend on the exact machine model.



Polynomial time is robust across standard computational models.

Closure Property

If an algorithm performs polynomial work and makes a constant number of calls to polynomial-time subroutines, then the total running time remains polynomial.

$$\mathcal{O}(n^a) + \mathcal{O}(n^b) = \mathcal{O}(n^{\max(a,b)})$$

$$\mathcal{O}(n^a) \cdot \mathcal{O}(n^b) = \mathcal{O}(n^{a+b})$$

2 Abstract Problems

Definition: Abstract Problem

An **abstract problem** Q is a binary relation between:

$I = \text{set of problem instances}$

and

$S = \text{set of problem solutions.}$

An instance may have one solution, many solutions, or no solution.

2.1 Example: Shortest Path

For the optimization problem SHORTEST-PATH:

instance = (G, u, v)

solution = a shortest path from u to v .

Shortest paths are not necessarily unique, so the same instance can be related to multiple valid solutions.

3 Decision Problems

Definition: Decision Problem

A **decision problem** is a problem whose output is either:

Yes or **No**

Equivalently, an abstract decision problem is a function:

$$Q : I \rightarrow \{0, 1\}.$$

NP-completeness focuses on decision problems. Many optimization problems can be converted into decision problems by adding a bound.

Decision Version of Shortest Path

The decision problem PATH asks:

Is there a path from u to v using at most k edges?

The encoded instance is:

$$i = \langle G, u, v, k \rangle$$

and the answer is:

$$\text{PATH}(i) = \begin{cases} 1, & \text{if such a path exists,} \\ 0, & \text{otherwise.} \end{cases}$$

Optimization to Decision

Instead of asking:

What is the minimum value?

we ask:

Is there a solution with value at most or at least k ?

This decision version is usually no harder than the original optimization problem and is the form used in NP-completeness theory.

4 Encodings

Definition: Encoding

An **encoding** of a set S of abstract objects is a function:

$$e : S \rightarrow \{0, 1\}^*$$

where $\{0, 1\}^*$ is the set of all finite binary strings.

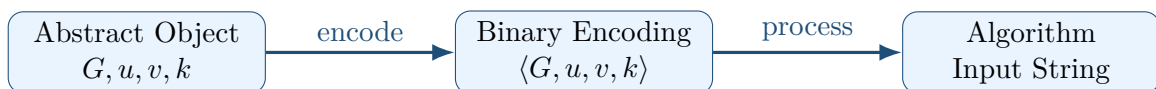
Computers do not directly manipulate abstract graphs, formulas, tuples, or programs. They manipulate binary strings. Therefore, every abstract instance must be represented by an encoding.

Examples

Natural numbers can be encoded in binary:

$$e(17) = 10001$$

Characters can be encoded using schemes such as ASCII. Compound objects such as ordered pairs, graphs, functions, and programs can be encoded by combining the encodings of their parts.



An algorithm receives the encoded string, not the abstract object itself.

5 Concrete Problems and the Class P

Definition: Concrete Problem

A **concrete problem** is a problem whose instance set is $\{0, 1\}^*$. That is, its inputs are binary strings.

An algorithm solves a concrete problem in time $\mathcal{O}(T(n))$ if, for every input string i of length

$$n = |i|,$$

the algorithm returns the correct answer in $\mathcal{O}(T(n))$ time.

The Class $\mathbf{P}$

$\mathbf{P} = \{L \subseteq \{0,1\}^* : L \text{ can be decided in time } \mathcal{O}(n^k) \text{ for some constant } k\}$.

Thus, $\mathbf{P}$ is the class of concrete decision problems solvable in polynomial time.

6 From Abstract Problems to Concrete Problems

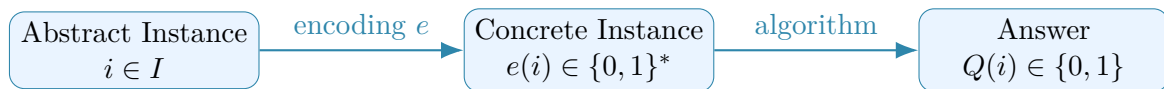
Let $Q : I \rightarrow \{0,1\}$ be an abstract decision problem, and let $e : I \rightarrow \{0,1\}^*$ be an encoding.

The encoding induces a concrete decision problem $e(Q)$:

$$e(Q)(e(i)) = Q(i).$$

Technical Convention

Some binary strings may not encode any meaningful instance. For convenience, these invalid strings can be assigned answer 0.



Encodings connect abstract mathematical problems to executable inputs.

7 Why Encoding Choice Matters

Important Point

Running time is measured as a function of input length, not as a function of the numeric value represented by the input.

Suppose an algorithm receives an integer k and runs in time $\Theta(k)$.

Encoding	Input Length	Time as a Function of Input Length
Unary	$n = k$	$\Theta(n)$, which is polynomial.
Binary	$n = \lfloor \log_2 k \rfloor + 1$	$\Theta(k) = \Theta(2^n)$, which is exponential.

Conclusion

The same algorithm can appear polynomial-time or exponential-time depending only on how the input is encoded.

This is why complexity theory normally avoids expensive or artificial encodings, such as unary encodings of large numerical values, unless they are explicitly part of the problem definition.

8 Polynomial-Time Computable Functions

Definition: Polynomial-Time Computable Function

A function

$$f : \{0, 1\}^* \rightarrow \{0, 1\}^*$$

is **polynomial-time computable** if there exists an algorithm that, for every input $x \in \{0, 1\}^*$, outputs $f(x)$ in polynomial time in $|x|$.

Such functions are used to translate one encoding into another without changing the problem's polynomial-time complexity.

9 Polynomially Related Encodings

Definition: Polynomially Related Encodings

Two encodings $e_1, e_2 : I \rightarrow \{0, 1\}^*$ are **polynomially related** if there exist polynomial-time computable functions f_{12} and f_{21} such that, for every instance $i \in I$,

$$f_{12}(e_1(i)) = e_2(i)$$

and

$$f_{21}(e_2(i)) = e_1(i).$$

In words, each encoding can be converted into the other in polynomial time.

Examples of Reasonable Encodings

Binary, decimal, base-3, and ASCII-style structured encodings are usually polynomially related. Therefore, changing between them does not change whether a problem is polynomial-time solvable.

10 Encoding Invariance Lemma

Lemma Statement

Let Q be an abstract decision problem on instance set I , and let e_1 and e_2 be polynomially related encodings of I . Then:

$$e_1(Q) \in \mathbf{P} \iff e_2(Q) \in \mathbf{P}.$$

Lemma 1 (Encoding Invariance). *If two encodings of the same abstract decision problem are polynomially related, then membership in $\mathbf{P}$ is independent of which encoding is used.*

Proof. It is enough to prove one direction, since the reverse direction is symmetric.

Assume $e_1(Q)$ can be solved in time $\mathcal{O}(n^k)$ for some constant k . Also assume that $e_1(i)$ can be computed from $e_2(i)$ in time $\mathcal{O}(n^c)$, where

$$n = |e_2(i)|.$$

To solve $e_2(Q)$ on input $e_2(i)$:

1. Convert $e_2(i)$ into $e_1(i)$.
2. Run the polynomial-time algorithm for $e_1(Q)$ on $e_1(i)$.

The conversion takes $\mathcal{O}(n^c)$ time, so the produced encoding has length

$$|e_1(i)| = \mathcal{O}(n^c).$$

The algorithm for $e_1(Q)$ then takes

$$\mathcal{O}(|e_1(i)|^k) = \mathcal{O}((n^c)^k) = \mathcal{O}(n^{ck}).$$

Thus the total running time is polynomial. Therefore $e_2(Q) \in \mathbf{P}$. □

11 Standard Encodings

Reasonable Standard Encoding

To discuss abstract problems without repeating encoding details, complexity theory assumes that instances are encoded in a concise, standard way.

A standard encoding usually has these properties:

- integers are encoded in a way polynomially related to binary;
- finite sets are encoded in a way polynomially related to lists of elements;
- tuples, graphs, formulas, and programs are built from these basic encodings;
- the notation $\langle X \rangle$ means the standard encoding of object X .

When to Be Careful

In most theoretical discussions, the exact standard encoding does not matter. However, in practical or unusual problems, the encoding may not be obvious. If the encoding changes the input length dramatically, it can change the apparent complexity of the problem.

12 Key Ideas

Term	Meaning
Polynomial time	Running time $\mathcal{O}(n^k)$ for some constant k .
P	The class of concrete decision problems solvable in polynomial time.
Abstract problem	A mathematical relation between instances and solutions.
Decision problem	A problem whose answer is Yes or No .
Encoding	A mapping from abstract objects to binary strings.
Concrete problem	A problem whose inputs are binary strings.
Polynomially related encodings	Encodings that can be translated into each other in polynomial time.
Standard encoding	A concise encoding convention used to avoid artificial complexity changes.

13 Quick Self-Check

Review Questions

1. Why is polynomial time considered a robust notion of tractability?
2. What is the difference between an abstract problem and a concrete problem?
3. Why must abstract problem instances be encoded before an algorithm can process them?
4. Why can unary and binary encodings give different apparent running times?
5. What does it mean for two encodings to be polynomially related?
6. Why does the encoding invariance lemma let us ignore most reasonable encoding choices?